

Model Selection

1 Masses, Decay Constants and Initial Conditions

This is to guide us how hyper-parameters are related to physical parameters in each models. In general, the matrix is defined at the Lagrangian level as

$$\mathcal{L} = \frac{1}{2}K_{ij}\partial_\mu\phi^i\partial^\mu\phi^j + \frac{1}{2}M_{ij}\phi^i\phi^j \quad (1)$$

$$= \frac{1}{2}\partial_\mu\boldsymbol{\phi}^T\mathbf{K}\partial_\mu\boldsymbol{\phi} + \frac{1}{2}\boldsymbol{\phi}^T\mathbf{M}\boldsymbol{\phi} \quad (2)$$

Then after the diagonalisation of the Kahler metric with the rotation matrix $\boldsymbol{\phi} = \mathbf{U}\boldsymbol{\phi}'$, the Lagrangian is in the form

$$\mathcal{L} = \frac{1}{2}\partial_\mu\boldsymbol{\phi}'^T\mathbf{U}^T\mathbf{K}\mathbf{U}\partial_\mu\boldsymbol{\phi}' + \frac{1}{2}\boldsymbol{\phi}'^T\mathbf{U}^T\mathbf{M}\mathbf{U}\boldsymbol{\phi}' \quad (3)$$

$$= \frac{1}{2}\partial_\mu\boldsymbol{\phi}'^T\mathbf{K}_D\partial_\mu\boldsymbol{\phi}' + \frac{1}{2}\boldsymbol{\phi}'^T\mathbf{U}^T\mathbf{M}\mathbf{U}\boldsymbol{\phi}' \quad (4)$$

This is the step that we define decay constants, $f_{eff}^i \equiv f_i$, as

$$\mathbf{K}_D = \text{diag}(f_1^2, \dots, f_n^2) \quad (5)$$

that is the effective decay constants squared are the eigenvalues of the Kahler matrix.

Then, rescale the fields with

$$\boldsymbol{\phi}' = \mathbf{K}_{reciprocal}\boldsymbol{\phi}'', \quad \mathbf{K}_{reciprocal} = \text{diag}\left(\frac{1}{\sqrt{K_{11}^D}}, \dots, \frac{1}{\sqrt{K_{nn}^D}}\right) \quad (6)$$

such that

$$\mathbf{K}_{reciprocal}^T\mathbf{K}_D\mathbf{K}_{reciprocal} = \mathbf{1} \quad (7)$$

and

$$\boldsymbol{\phi} = \mathbf{U}\mathbf{K}_{reciprocal}\boldsymbol{\phi}'' \quad (8)$$

At this stage the Lagrangian becomes

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi''^T \partial_\mu \phi'' + \frac{1}{2} \phi''^T \mathbf{K}_{reciprocal}^T \mathbf{U}^T \mathbf{M} \mathbf{U} \mathbf{K}_{reciprocal} \phi'' \quad (9)$$

Before shift symmetry is broken, the canonically normalised axions are randomly distributed everywhere in field space. So we give the initial conditions for ϕ'' and $\partial_\mu \phi''$ at this step as

$$\phi''_{in} = \begin{pmatrix} \theta_{in}^1 f_1 \\ \vdots \\ \theta_{in}^n f_n \end{pmatrix}, \quad \dot{\phi}''_{in} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \quad (10)$$

where θ_{in}^i is uniformly distributed from $-\pi$ to π

$$\theta_{in}^i \in [-\pi, \pi] \quad (11)$$

Finally we apply the rotation matrix $\phi'' = \mathbf{V} \tilde{\phi}$ such that the final mass matrix is diagonal.

$$\mathbf{V}^T \mathbf{K}_{reciprocal}^T \mathbf{U}^T \mathbf{M} \mathbf{U} \mathbf{K}_{reciprocal} \mathbf{V} = \mathbf{M}_D \quad (12)$$

and the Lagrangian is nice and simple

$$\mathcal{L} = \frac{1}{2} \partial_\mu \tilde{\phi}^T \partial_\mu \tilde{\phi} + \frac{1}{2} \tilde{\phi}^T \mathbf{M}_D \tilde{\phi} \quad (13)$$

1.1 Summary

In the basis of $\tilde{\phi}$ the mass is can be found by

$$\boxed{\mathbf{M}_D = \mathbf{V}^T \mathbf{K}_{reciprocal}^T \mathbf{U}^T \mathbf{M} \mathbf{U} \mathbf{K}_{reciprocal} \mathbf{V}} \quad (14)$$

where $\mathbf{U}$ and $\mathbf{V}$ are diagonalisation matrices of $\mathbf{K}$ and $\mathbf{K}_{reciprocal}^T \mathbf{U}^T \mathbf{M} \mathbf{U} \mathbf{K}_{reciprocal}$ respectively and

$$\boxed{\mathbf{K}_{reciprocal} = \text{diag}\left(\frac{1}{\sqrt{K_{11}^D}}, \dots, \frac{1}{\sqrt{K_{nn}^D}}\right)} \quad (15)$$

The effective decay constant can be found from

$$\boxed{\mathbf{K}_D = \text{diag}(f_1^2, \dots, f_n^2)} \quad (16)$$

The initial conditions in the basis of $\tilde{\phi}$ is written as

$$\boxed{\tilde{\phi}_{in} = \mathbf{V}^T \phi''_{in} = \mathbf{V}^T \begin{pmatrix} \theta_{in}^1 f_1 \\ \vdots \\ \theta_{in}^n f_n \end{pmatrix}} \quad (17)$$

2 Models

We need two sets of mean and variance for each model, i.e., (a_0, σ_a) and (b_0, σ_b) . The exception is made in MT model where the hyper parameters of K is (a_0, a_1, a_2)

2.1 Easter-McAllister model (EM)

In the code the tag name of the model is EM.

The Kahler matrix is assumed to be

$$K = \mathbf{1} \quad (18)$$

The potential matrix is assumed to be

$$M = B \times B^T, \quad B = \begin{pmatrix} r_{11} & \dots & r_{1n} \\ \vdots & \ddots & \vdots \\ r_{n1} & \dots & r_{nn} \end{pmatrix} \quad (19)$$

where r_{ij} are i.i.d values drawn from the normal distribution with mean 0 and variance σ_b^2 . B is a $(N + P)N$ rectangular matrix. This will give the marchenko pastur density function for the eigenvalues of M .

2.2 Logflat model (LF)

Parameterisation used so far is log normal NOT log flat. I have changed to log flat. Also minus sign is now correct.

In the code the tag name of the model is LF.

The Kahler matrix is assumed to be

$$K = A \times A^T, \quad A = \begin{pmatrix} e_{11} & \dots & e_{1n} \\ \vdots & \ddots & \vdots \\ e_{n1} & \dots & e_{nn} \end{pmatrix} \quad (20)$$

where $e_{ij} = e^{r_{ij}}$ and r_{ij} are uniform random number ranging from

$$r_{ij} \in [\ln a_0 - \sigma_a, \ln a_0 + \sigma_a] \quad (21)$$

$$r_{ij} \in U[k_{\min}, k_{\max}] \quad (22)$$

The potential matrix is assumed to be

$$M = B \times B^T, \quad B = \begin{pmatrix} e_{11} & \dots & e_{1n} \\ \vdots & \ddots & \vdots \\ e_{n1} & \dots & e_{nn} \end{pmatrix} \quad (23)$$

where $e_{ij} = e^{r_{ij}}$ and r_{ij} are uniform random number ranging from

$$r_{ij} \in [\ln b_0 - \sigma_b, \ln b_0 + \sigma_b] \quad (24)$$

$$r_{ij} \in U[m_{\min}, m_{\max}] \quad (25)$$

2.3 M-theory inspired model (MT)

In the code the tag name of the model is MT.

The Kahler matrix is assumed to be

$$K = A \times A^T, \quad A = a_0 \begin{pmatrix} \frac{1}{s_{11}} & \dots & \frac{1}{s_{1n}} \\ \vdots & \ddots & \vdots \\ \frac{1}{s_{n1}} & \dots & \frac{1}{s_{nn}} \end{pmatrix} \quad (26)$$

where s_i are uniform random numbers ranging from

$$s_i \in [a_1, a_2] \quad (27)$$

The potential matrix is assumed to be

$$M = B \times B^T, \quad B = \begin{pmatrix} e_{11} & \dots & e_{1n} \\ \vdots & \ddots & \vdots \\ e_{n1} & \dots & e_{nn} \end{pmatrix} \quad (28)$$

where $e_{ij} = e^{r_{ij}}$ and r_{ij} are uniform random number ranging from

$$r_{ij} \in [\ln b_0 - \sigma_b, \ln b_0 + \sigma_b] \quad (29)$$